PLASTIC THEORY OF BENDING

PLASTIC THEORY OF BENDING

- In any particular loaded beam, if the load system were increased gradually, yielding would first occur at the extreme fibres of the" weakest" section .These fibres are then said to be in the *plastic* state, and further increase in loading will bring about a considerable increase in strain and hence deflection(collapse) at that section of the beam(plastic hinge).

fully elastic⟶elastoplastic⟶fully plastic⟶collapse

Moment of resistance

- Moment of resistance in the elastic region

$$M_e = \frac{\sigma I}{c}$$

- Moment of resistance in the plastic region

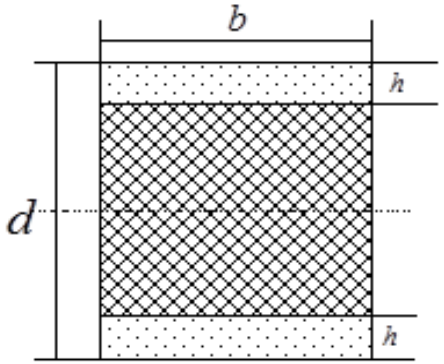
$$M_p = Fx = \sigma A x$$

- Moment of resistance in the elastoplastic region

$$M = M_e + M_p = \frac{\sigma I}{c} + \sigma A x$$

Example

Consider the rectangular section in elastoplastic state shown, h is the depth of plastic penetration



$$M = M_e + M_p = \frac{\sigma I}{c} + \sigma A x$$

$$I = \frac{b(d-2h)^3}{12}, c = \frac{d-2h}{2}$$

$$A = 2hb, x = \frac{d-2h}{2} + \frac{h}{2} = \frac{d-h}{2}$$

$$M = \sigma_y \left[\frac{b(d-2h)^2}{6} + hb(d-h) \right]$$

In the fully plastic state, h = d/2

$$\therefore \mathrm{M_p} = \sigma_y \frac{bd^2}{4}$$

In a fully elastic state, h = 0;

$$\mathrm{M_e} = \sigma_y \frac{bd^2}{6}$$

The ratio $\mathrm{M_p}/\mathrm{M_e}$ is called the shape factor S, since it depends only on the shape of the cross-section, i.e. for a rectangular section,

$$\frac{M_p}{M_e} = 1.5$$

Example 1

- A steel bar of rectangular section 72 mm by 30 mm is used as a simply supported beam on a span of 1.2 m and loaded at mid-span. If the yield stress is 280 N/mm^2 and the long edges of the section are vertical, find the load when yielding first occurs.
- Assuming that a further increase in load causes yielding to spread inwards towards the neutral axis, with the stress in the yielded part remaining at 280 N/mm^2, find the load required to cause yielding for a depth of 12 mm at the top and bottom of the section at mid-span, and find the length of beam over which yielding has occurred.

Example 1 cont'd…

If W_y is the load at first yield, then

$$M_y = \left(\frac{bd^2}{6}\right)\sigma_y = \frac{(30)(72)^2(280)}{6}$$

$$= 300W_y \Rightarrow W_y = 24.2kN$$

In the elastoplastic state

$$M = \tfrac{1}{6}\sigma_y b(d-2h)^2 + \sigma_y bh(d-h)$$

$$= \tfrac{1}{6}(280)(30)(48)^2 + (280)(30)(12)(60)$$

$$M = 928 \text{ x}10^4 \text{ Nmm}$$

$$= 300W = 9280000 \Rightarrow W = 31\,kN$$

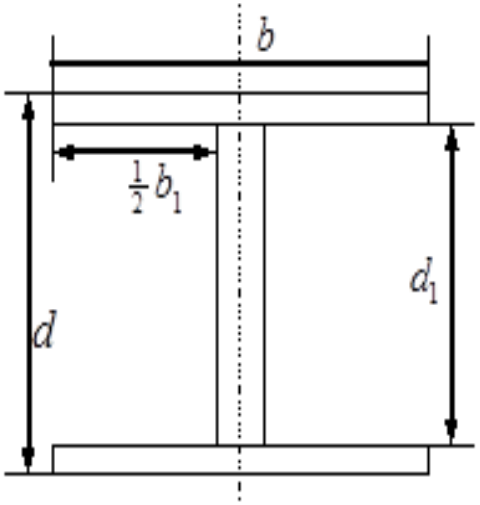
At first yield the moment of resistance is, $W_y(0.3)$, that is 7.26 kNm and if this occurs at a distance x from either end under the central load W, then $\tfrac{1}{2}Wx = 7.26$

$$\Rightarrow x = 0.468m$$

The length of the beam over which yielding occurs is

$$\delta = 1.2 - 2x = 0.264\,m$$

I-Section



Consider an I - beam when yielding is about to occur at the extreme fiber, we have

$$M_y = \sigma_y \left(\frac{bd^3}{12} - \frac{b_1 d_1^3}{12} \right) \left(\frac{2}{d} \right)$$

In a fully plastic condition, we have

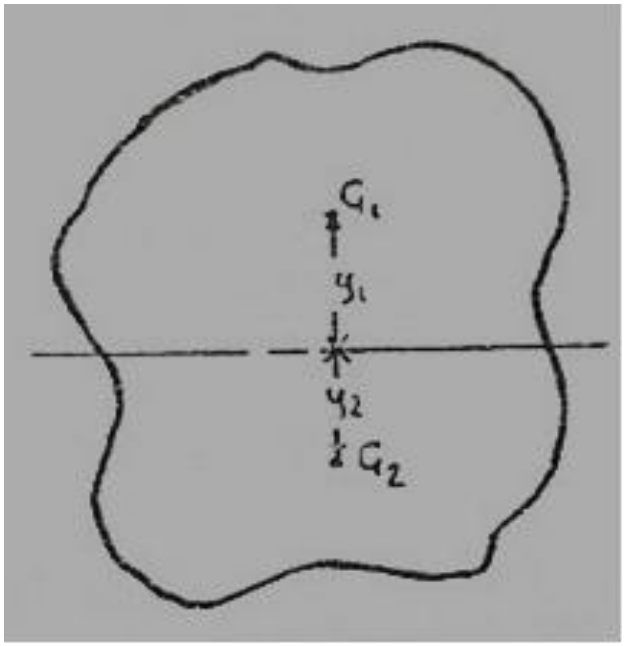
$$M_p = \sigma_y \left(\frac{bd^2}{4} - \frac{b_1 d_1^2}{4} \right)$$

The shape factor is

$$S = \frac{M_p}{M_y} = 1.5d \left(\frac{bd^2 - b_1 d_1^2}{bd^3 - b_1 d_1^3} \right)$$

General Case

If *A* is the total area of cross-section shown below, then it is clear that for pure bending in the fully plastic state the "neutral axis" must divide the area into equal halves. If the centroids of these halves are G_1 and G_2 at a distance $y_1 + y_2$ apart, then



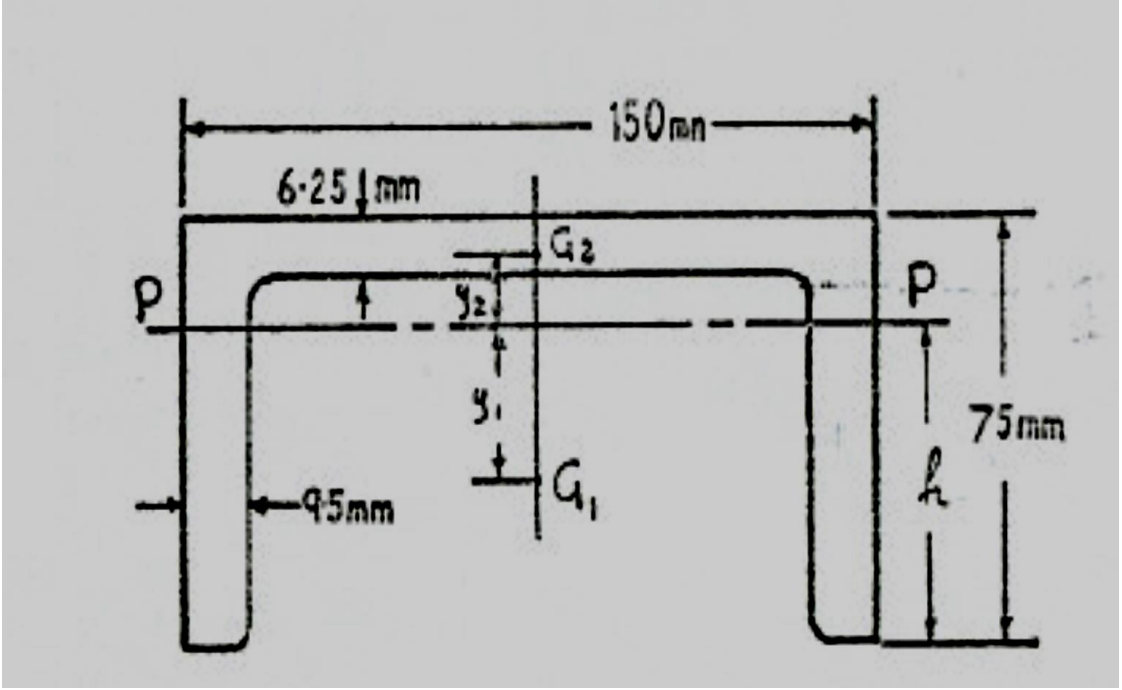
$$M_y = Z\sigma_y, \text{ where Z is the section modulus}$$

$$S = \frac{M_p}{M_y} = \frac{A(y_1 + y_2)}{2Z}$$

Example 3

Find the shape factor for a 150 mm by 75 mm channel in puree bending with the plane of bending perpendicular to the web of the channel.

The dimensions are shown, and A=2300 mm^2 *and* Z =12,000 mm^3.



Example 3 cont'd…

Let PP be the neutral axis under fully plastic conditions, then this divides the total area equally.

Assuming all corners are 'square', then

$$2(9.5)h = 150(6.25) + 2(9.5)(75 - h - 6.25)$$

$$\Rightarrow h = 59mm$$

The centroids of the two areas on either side of PP are G_1 and G_2 at distances y_1 and y_2, where

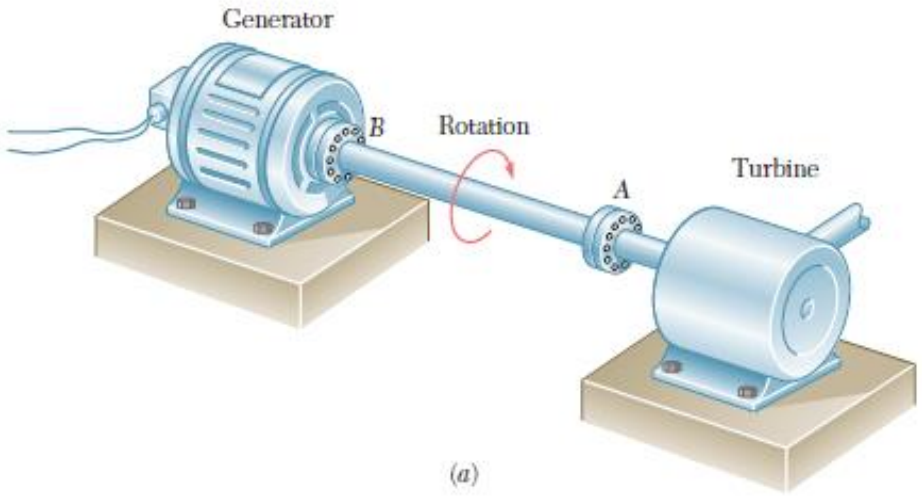
$$y_1 = h/2 = 29.5mm$$

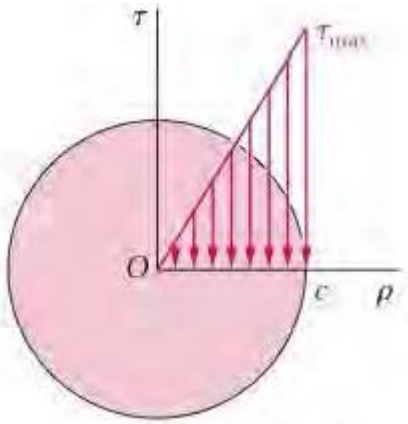
$$y_2 = \frac{150(6.25)(16 - 3.125) + (2)(9.5)(9.75)(4.875)}{150(6.25) + 2(9.5)(9.75)}$$

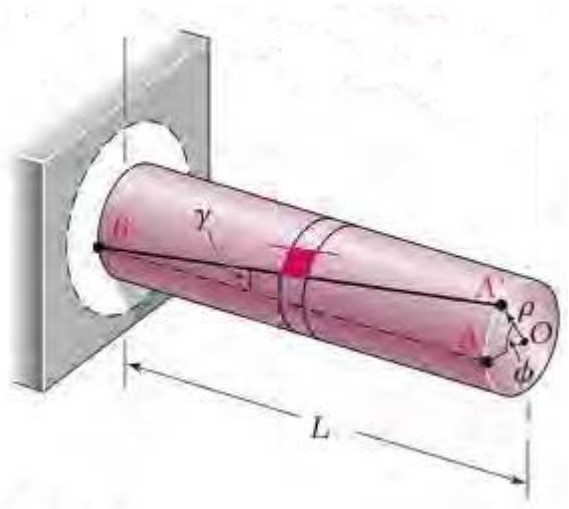
$$= 11.8mm$$

$$S = \frac{A(y_1 + y_2)}{2Z} = \frac{2300(41.3)}{2(21000)} = 2.26$$

Torsion in a circular shaft







$$AA' = \rho\emptyset = L\gamma$$

$$T = \int_0^c r dF = \int_0^c r\tau dA$$

$$rT = \int_0^c r^2 dA$$

$$rT = \tau J; \quad J = \int r^2 dA$$

$$\tau = \frac{Tr}{J}; \quad J = \frac{\pi d^4}{32} = \frac{\pi r^4}{2}$$

Torsion under Plastic Condition

Assumptions in the* Plastic *Theory

1.A plane cross section of the shaft remains plane when in the plastic state.

2.Radius remains straight.

When the shaft has a torque T applied in the elastic range, the shear stress increases from zero at the axis to a maximum at the surface of the shaft, thus

$$T = \frac{\tau \pi r^3}{2}$$

When the torque is increased beyond this value, the plastic zone advances towards the middle for solid.When the shear stress at the surface reaches the yielding value,

$$T_y = \frac{\tau_y \pi r^3}{2}$$

Torsion under Plastic Condition

The torque carried by the elastic core is

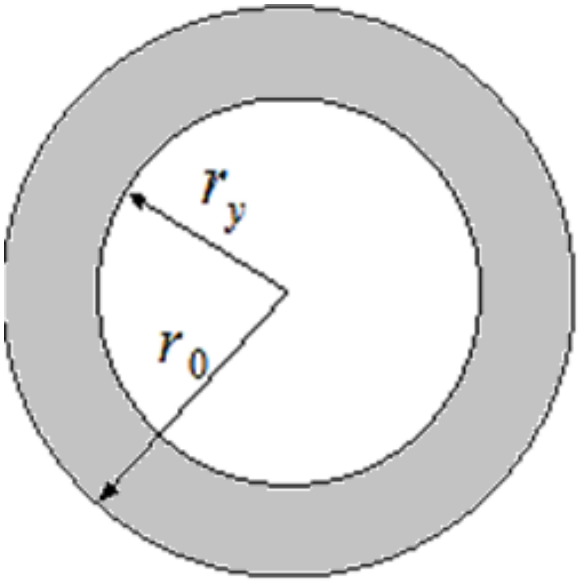
$T_e = \dfrac{\tau_y \pi r_y^3}{2}$, r_y is the interface radius.

The torque carried by the elasto - plastic zone is

The total torque is

$$T = T_e + T_p = \tfrac{1}{2}\pi\tau_y r_y^3 + \tfrac{2}{3}\pi\tau_y\left(r_0^3 - r_y^3\right)$$

$$\therefore T = \frac{2}{3}\pi\tau_y r_0^3\left(1 - \frac{r_y^3}{4r_0^3}\right)$$



When fully plastic, $r_y = 0$, thus

$$T_p = \frac{2}{3}\pi\tau_y r_0^3$$

and the ratio is $\tau_p / \tau_y = 4/3$

Torsion under Plastic Condition

When the fibres at the outer surface of the shaft are about to become plastic, the angleof twist of the shaft is given by; $\theta_y = \frac{\tau_y L}{G r_0}$

L the length of the shaft
G Modulus of Rigidity

When the shaft is in the elastoplastic condition, the angle of twist of the elastic core is given by

$$\theta = \frac{\tau_y L}{G r_y}$$

Torsion under Plastic Condition

Since dividing, we have assumed that radii remain straight, then ; $\frac{\theta_y}{\theta} = \frac{r_y}{r_0}$ It is evident that as the shaft approaches the fully plastic state, the angle of twist tends to be infinity.

It is evident that as the shaft approaches the fully plastic state, the angle of twist tends to be infinity.

$$T = \frac{2}{3}\pi\tau_y r_0^3\left(1 - \tfrac{1}{4}\left(\frac{\theta_y}{\theta}\right)^3\right)$$

Example

A mild steel shaft 40 mm in diameter and 250 mm in length is subjected to an overload torque of 1800 Nm which caused sheer yielding of the shaft. Determine the radial depth to which plasticity has penetrated and the angle of twist.

Solution

$$T = \frac{2}{3}\pi\tau_y r_0^3\left(1-\tfrac{1}{4}\left(\frac{\theta_y}{\theta}\right)^3\right)$$

$$1800 = \frac{2}{3}\pi\left(120.10^6\right)\left(0.02\right)^3\left[1-\frac{r_y^3}{4\left(0.02\right)^3}\right]$$

From which $r_y = 15\,mm$

Hence depth of plastic deformation is 5mm.

The shear strain at ry $= 15$mm is

$$\gamma = \frac{r_y\theta}{L} = \frac{\tau_y}{G} = \frac{120\left(10^6\right)}{80\left(10^9\right)} = 1.5x10^{-3}$$

$$\Rightarrow \theta = \frac{\gamma L}{r_y} = \frac{\left(0.0015\right)\left(0.25\right)}{0.015} = 0.025\,rad = 1.43^0$$

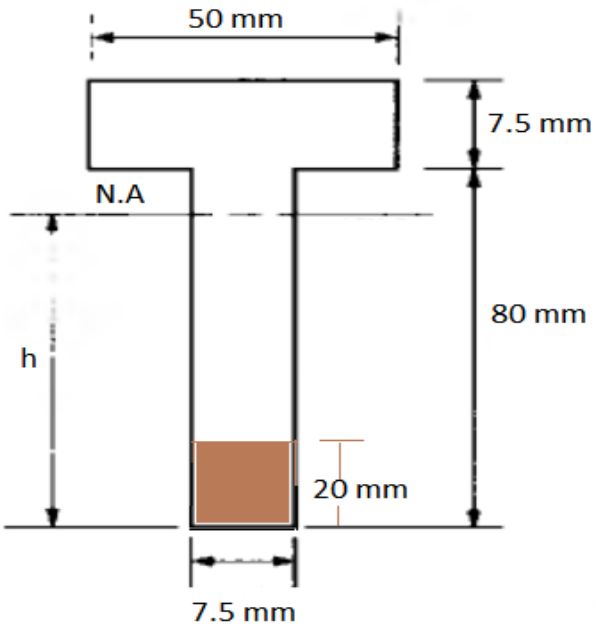
Assignment 3

1. The figure below shows the section of a beam which is subjected to a bending moment of such magnitude that yielding occurs at the lower part of the web over a depth of 2 cm. The yield stress of 280 N/mm^2 may be assumed constant over the yielding area, while over the remainder of the section the stress is proportional to the distance from the neutral axis. Determine:

   A. the position of the neutral axis

   B. the stress at the top of the section

   C. the moment of resistance of the section



Assignment 3 cont'd…

2. A steel shaft of external diameter 36 mm and internal diameter 20 mm, is subjected to an over load torque of 1.35 kN-m, which caused shear yield. If the shaft is 450 mm long. Calculate the depth of penetration and the angle of twist.

   $\tau_y = 125\ N/mm^2$ and G=80 000 N/mm^2